Santhameena P

Data Science Enthusiast

meenamanikandan3011@gmail.com | 7867950911 | Tiruppur, Tamilnadu | LinkedIn | Github

PROFILE

Experienced Technical Instructor with 1.8 years, skilled in Cloud Computing, Embedded Systems, data analysis, and dedicated to extracting valuable business insights. Transitioning with a solid foundation in **Data Science, Machine Learning, Natural Language Processing and Generative-AI** committed to ongoing learning and applying advanced data science techniques to drive business growth. Eager to contribute to data science team with innovative solutions

SKILLS

Python | Data Visualization | Machine Learning | Deep Learning | Power BI | SQL | LLM | Hugging Face | Flowise AI | AWS Cloud | NLP | Streamlit

PROJECTS

Multiple Disease Prediction System
Key Skills: Python, Machine Learning, Streamlit, AWS EC2
• Developed machine learning algorithms like **Logistic Regression and Random Forest** to predict **Kidney, Liver, and Parkinson's diseases** using healthcare datasets
• Performed data preprocessing, feature scaling like **MinMaxScalar**, **Cross-Validation** to ensure robustness and model evaluation to improve prediction accuracy
• Built and deployed an interactive pickling file in **Streamlit web application** for real-time disease prediction
• Hosted the application on **AWS EC2**, enabling scalable and accessible healthcare predictions

NLP Sentiment Analysis – AI Echo
Key Skills: Python, NLP, BERT, Scikit-learn, Streamlit
• Built a sentiment analysis system to classify user reviews using **NLP techniques and machine learning models**
• Performed text preprocessing, EDA, and **feature extraction (TF-IDF)** for model improvement
• Leveraged pretrained **bert-based-uncased** Transformer based **BERT model** to enhance sentiment classification accuracy
• Developed a **Streamlit dashboard** for real-time sentiment prediction and visualization

Policy Insight & Impact Analyzer (PPIIA)
Key Skills: Python, NLP, LLM (Llama 3), Prompt Engineering, Streamlit, PyMuPDF, API Integration, Flowise AI
• Developed an **end-to-end AI-powered policy analysis system** to process Indian Parliament bills and generate structured insights using LLMs
• Built a scalable pipeline for **PDF text extraction (PyMuPDF), preprocessing, chunking, and summarization**, enabling efficient handling of large documents
• Designed advanced **prompt engineering workflows** to produce multi-dimensional analysis including sector classification, objectives, risks, and impact (short/medium/long-term)
• Implemented an interactive **Streamlit web application with chatbot functionality**, allowing users to query bills and download summaries as PDF, Integrated **Flowise AI workflows** for efficient PDF processing and automated insights generation

PROFESSIONAL EXPERIENCE

Technical Instructor — Jun 2016 – Nov 2017 | Coimbatore
AdroIT Technologies
- Trained students in C, C++, Java, Python, MySQL, Cloud Computing and Embedded Systems
- Delivered IBM-authorized tool training like Rational Rhapsody, Bluemix, Rational Application Developer
- Conducted C++ programming sessions in Chandigarh University (IBM Program)

EDUCATION

IIT-M Advanced Programming Professional & Master Data Science — Sep 2025 – Dec 2025 | Chennai
IIT-M GUVI

Bachelor of Engineering in Electronics and Communication Eng. — Jun 2013 – Apr 2016 | Tirupur
Angel College of Engineering and Technology

CERTIFICATES

- Advanced Professional Certification in Data Science from GUVI GREEK NETWORK PRIVATE LIMITED